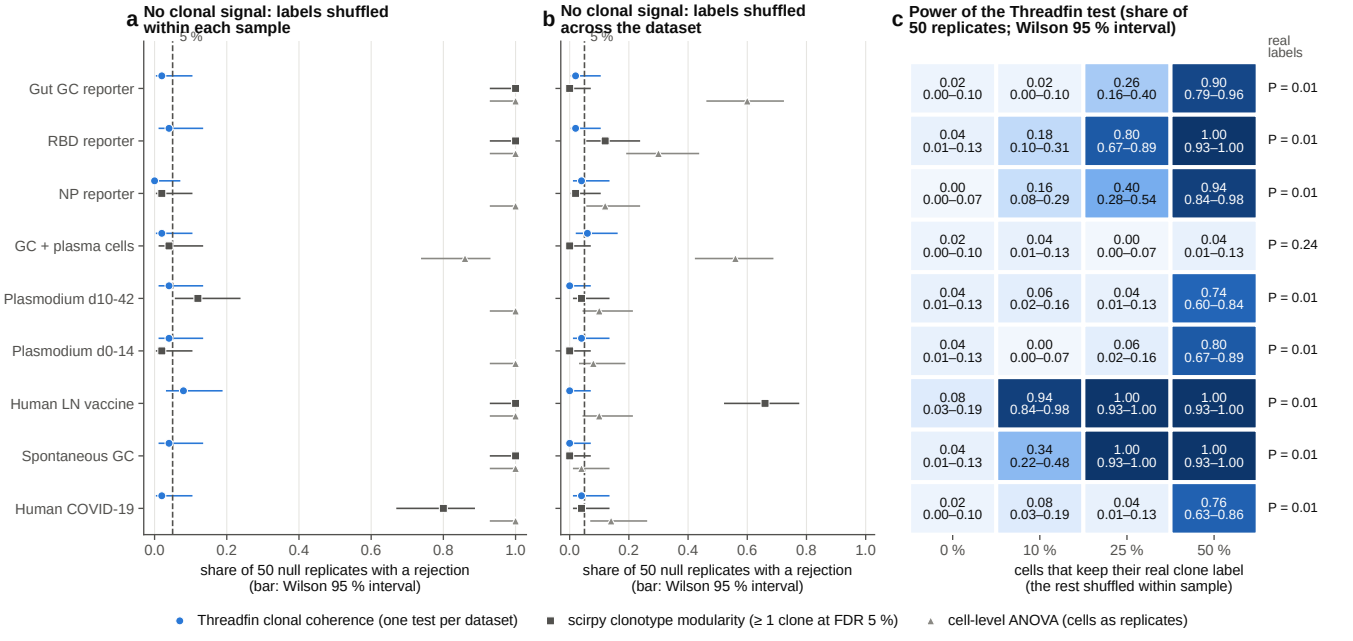
a
No clonal signal: labels shuffled within each sample
5 %
Gut GC reporter
RBD reporter
NP reporter
GC + plasma cells
Plasmodium d10-42
Plasmodium d0-14
Human LN vaccine
Spontaneous GC
Human COVID-19
0.0
0.2
0.4
0.6
0.8
1.0
share of 50 null replicates with a rejection
(bar: Wilson 95 % interval)
b
No clonal signal: labels shuffled across the dataset
5 %
0.0
0.2
0.4
0.6
0.8
1.0
share of 50 null replicates with a rejection
(bar: Wilson 95 % interval)
c
Power of the Threadfin test (share of 50 replicates; Wilson 95 % interval)
real labels
0.02 0.00–0.10
0.02 0.00–0.10
0.26 0.16–0.40
0.90 0.79–0.96
P = 0.01
0.04 0.01–0.13
0.18 0.10–0.31
0.80 0.67–0.89
1.00 0.93–1.00
P = 0.01
0.00 0.00–0.07
0.16 0.08–0.29
0.40 0.28–0.54
0.94 0.84–0.98
P = 0.01
0.02 0.00–0.10
0.04 0.01–0.13
0.00 0.00–0.07
0.04 0.01–0.13
P = 0.24
0.04 0.01–0.13
0.06 0.02–0.16
0.04 0.01–0.13
0.74 0.60–0.84
P = 0.01
0.04 0.01–0.13
0.00 0.00–0.07
0.06 0.02–0.16
0.80 0.67–0.89
P = 0.01
0.08 0.03–0.19
0.94 0.84–0.98
1.00 0.93–1.00
1.00 0.93–1.00
P = 0.01
0.04 0.01–0.13
0.34 0.22–0.48
1.00 0.93–1.00
1.00 0.93–1.00
P = 0.01
0.02 0.00–0.10
0.08 0.03–0.19
0.04 0.01–0.13
0.76 0.63–0.86
P = 0.01
0 %
10 %
25 %
50 %
cells that keep their real clone label
(the rest shuffled within sample)
Threadfin clonal coherence (one test per dataset)
scirpy clonotype modularity (≥ 1 clone at FDR 5 %)
cell-level ANOVA (cells as replicates)